

Bachelor in Computer Applications

Course Title: **Numerical Methods**

Code No: Semester: **3**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Define true error and relative error. Derive the bisection method for solving non-linear equation and using this method solve $(2x^3 - 2x - 5)$ with initial $(x_0 = 1)$ and $(x_1 = 2)$. Calculate upto (10^{th}) iteration.
2. What are the applications of interpolation? Differentiate between interpolation and regression. Consider the following data points estimate the $(f(10))$ using Lagrange's interpolation.
x56911y13141516
3. What do you mean by numerical integration? Write any one application of numerical integration. Write an algorithm and c program to implement multi-segment trapezoidal rule.
4. Construct the divided difference table for the following data value and find first and second order derivative at $(x = 2)$.
x124810y0152127
5. Why it is better to use composite Simpson's $(\frac{3}{8})$ rule instead of Simpson's $(\frac{3}{8})$ rule? Find the value of integration for following data set using Simpson's $(\frac{3}{8})$ rule.
x0123456y10.50.20.10.05880.03850.027
6. What is least squares method of fitting a function? Fit the second order polynomial for the following data values.
x1234567y2678101215
7. Solve the following system of linear equations using Gaussian elimination method.
 $(2x + 2y + z = 12)$
 $(3x + 2y + 2z = 8)$
 $(5x + 10y - 8z = 10)$
8. Solve the following system of linear equations using Gauss-Seidal method.
 $(10x + y + z = 12)$
 $(2x + 10y + z = 13)$
 $(2x + 2y + 10z = 14)$
9. Define eigen value and eigen vector. Explain how shooting method is used to solve boundary value problem.
10. Find the approximate value of (y) when $(x = 0.6)$ of $(\frac{dy}{dx} = 1 - 2xy)$, given that $(y = 0)$ when $(x = 0)$ with $(h = 0.2)$ using Heun's method.
11. Consider a steel plate of size $24\text{cm} \times 24\text{cm}$. If two of the opposite sides are held at 100 degree Celsius and the other two opposite sides at 0 degree Celsius, find the steady state temperatures of interior points, assuming a grid size of $8\text{cm} \times 8\text{cm}$.
12. Write an algorithm for Horner's method. Evaluate the polynomial $(f(x) = x^4 + 3x^3 + 5x^2 + 7x + 9)$ at $(x = 2)$ by using Horner's method.

